

and more are finally becoming part of the mainstream.

While phone makers still remain undecided on the design language of the next generation of smartphones, one thing is clear, they won't be anything like the ones we carry in our pockets today. The primary reason for that will be the advancements in foldable display technology.

Foldable displays are set to forever change the form factors and design languages device makers adopt to craft new smartphones. Unlike the current generation devices that are bound to end up as slabs of metal and glass, the smartphone of the future is set to be a device that will be able

advancements we've seen in silicon technology over the past two decades. These advancements have also ensured that there is now a solid base for chip makers to work on, and take mobile computing to the next level in the coming years.

While Moore's Law dictates that the growth in transistor density within processors should double every two years, in reality that may not be the case as we've already reached a point in processing technology where it's not feasible to maintain such growth. However, that does not mean we'll not see major leaps in processing power of processors in the coming years.

ARM recently announced its next-generation CPU and GPU core designs. These are expected to be market ready by early 2022 and power processing units inside Android flagship chipsets later in the year. As per the information revealed by the chip-maker, the new CPU and GPU cores – Cortex-X2, Cortex-A710 CPUs, Cortex-A510 and Mali-G710 GPU

– are expected to come with a wide assortment of new features and a brand new architecture as well.

Chips based on cores using the ARMv9 architecture are being promised to bring massive jumps in performance and also add additional security as well as AI features. In fact, ARM says that a CPU cluster made up of its new cores – a single Cortex-X2, three Cortex-A710 cores, and four Cortex-A510 cores – should be bring improvements across the board as it will offer 30 per cent peak performance and about 30 per cent better overall efficiency compared to CPU configurations found inside current generation flagship chipsets powering smartphones.

Although ARM hasn't suggested the idea yet, we can also see more ambitious CPU configurations based on ARMv9 cores in the future that could take mobile computing to newer heights in the years to come.

5G REVOLUTION

Over the past two decades, network standards have slowly improved. From the roll-out of the first 3G network in Japan in 2001 to the on-going deployment of 5G networks across the globe in 2021, telecom networks have come a long way.

Compared to what the modest previous, and current generation networks can offer, next generation 5G networks promise improved connectivity, data speeds and latency. This is expected to make smartphones in the next few years truly wireless as they will cease to be tied down by the end user's need for fast Wi-Fi networks for their data needs.

5G wireless technology has a lot of benefits, with support for high data speeds definitely being one of them. These networks can deliver multi-Gbps peak data speeds (up to 20-Gigabits per second) at extremely high stability levels for more uniform user experience. However, that's not the only benefit of 5G. These networks are also designed to provide 100x more traffic capacity and network efficiency when compared to 4G.

Another major benefit that 5G will bring over current generation 4G networks will be the promise of 10x decrease in end-to-end latency. This will not only improve the performance of tasks such as video calling, and streaming of content, but also video game streaming. Additionally, 5G technology will also help bring to life immersive experiences such as VR and AR.

But what does 5G technology have to do with shaping the future of smartphone experiences going forward? Well, the short answer to that question is: EVERYTHING.

Much like how the roll-out of 3G networks in 2001 changed the way we started using our phones, roll-out of 5G networks in the coming months and years will change not only the way we interact with our phones, but possibly the very hardware that goes inside them. As such, 5G technology with its endless possibilities will completely revolutionise the smartphone industry itself.



Samsung foldable smartphone

to adopt a clamshell, book or even a rollable scroll like form factor. Interestingly, these are form factors we've already seen device makers work on for years, with the first and even second generation devices already available in markets.

The foldable display revolution will, however, not just stop with these form factors. Recent reports suggest there could be even more ambitiously designed phones in the near future, with Samsung even tipped to launch a triple-folding phone/tablet in the market as early as next year. This device will supposedly have two hinges and three folding segments.

COMPUTING POWER

Unlike the first smartphones, the modern-day smartphone does not compete with feature phones. Instead, it goes up against full-fledged PCs these days. This has largely been made possible because of the rapid